Project Documentation

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Bansilal Ramnath Agarwal Charitable Trust's

Vishwakarma Institute of Technology, Pune

(An autonomous Institute of Savitribai Phule Pune University)

LUMEN - Localized Unified Medical Engine for triage

Team LUMEN

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Theme:

AI for Societal Good + Learn Smarter: AI in Education + Open Innovation

Prototype website:

Github repo: [github]

2. Problem Statement

How do we ensure that rural women and families get timely, affordable, and reliable healthcare?

Why are preventable deaths still common in villages despite government schemes and technology progress?

India's rural healthcare system faces **critical gaps** that result in preventable deaths, untreated conditions, and rising costs.

- **Limited Access & Workforce Shortage**
  Over **900 million rural residents** (65% of the population) face inadequate infrastructure, with **16% fewer PHCs** and **50% fewer CHCs** than required. Shortages are severe: **8% PHCs lack doctors**, **38% lack lab technicians**, and workforce density is **20.6 per 10,000**, far below the WHO norm of 44.5.[DocBox] [BriefPMC] [BioMed]

- **Emergency Care Deficiencies**
  Snakebites alone cause **58,000 deaths annually**, 70% in rural areas where delays and lack of first-aid knowledge prevail. Many victims first turn to traditional healers, worsening outcomes.[PMCeLifeWikipedia] [The Times of India+1]

- **Women's Health & Menstrual Hygiene Gaps**
  Disorders like **PCOS (6–10% prevalence)** and endometriosis remain underdiagnosed, with low awareness in rural India. [PMC] [Reproductive Health Journal] [ref] [ref]
  Only **42–43% of adolescent girls** use hygienic menstrual products; poor practices increase risk of infections. [BMC Public Health] [ref]
  Stigma and taboos limit discussion and treatment, while awareness of government schemes like **MHS** and **Jan Aushadhi** remains low, adding travel and wage-loss costs for women. [ref] [ref]

These gaps result in **avoidable morbidity, mortality, and economic strain**, while existing policies remain fragmented and underutilized. A **holistic, context-aware intervention** is urgently needed.

3. Proposed Solution – LUMEN

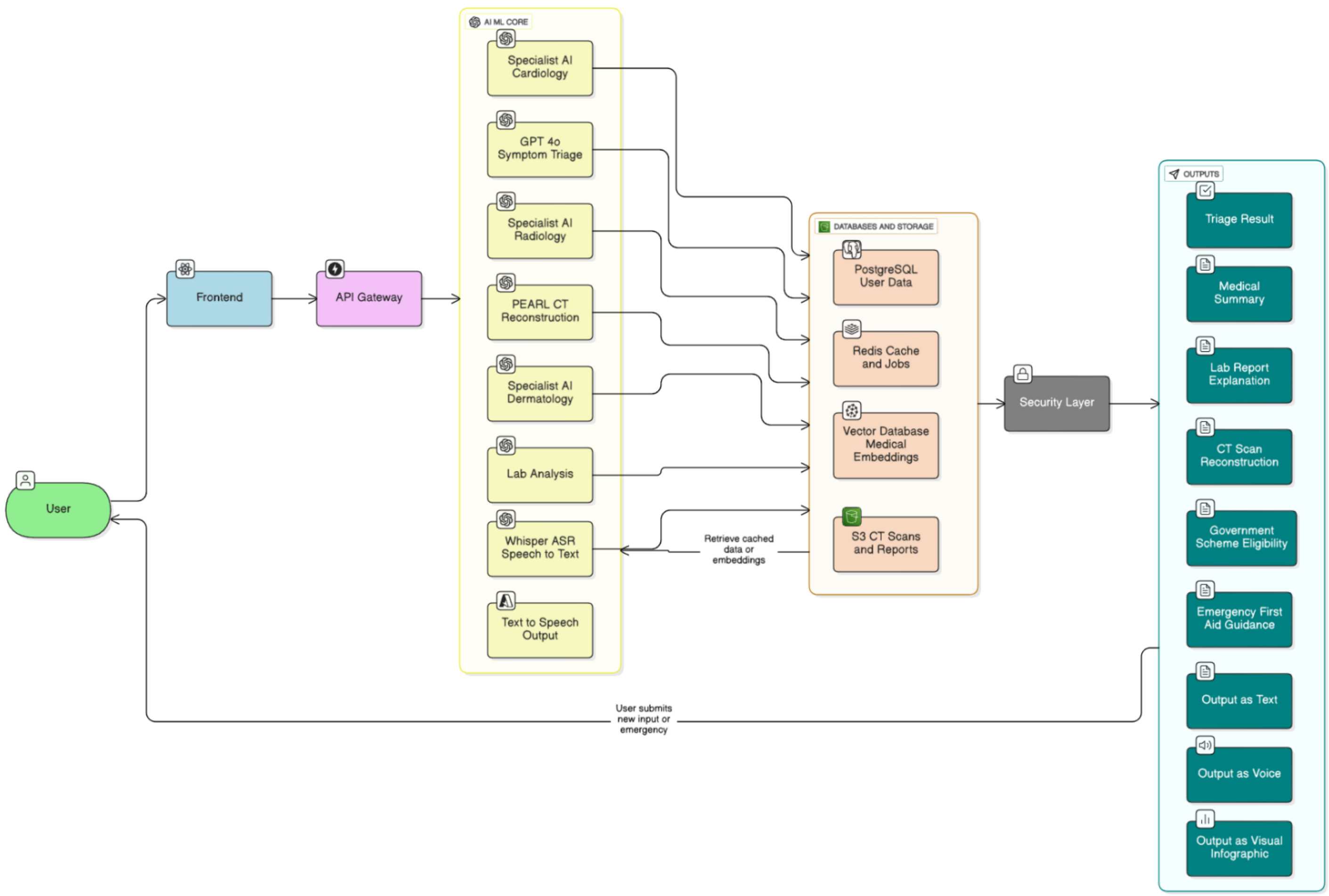

Diagram 1: System user flow diagram - “layman's terms diagram”

1. **User**
   - Patients give inputs: symptoms (text/img/doc), lab/CT uploads, or emergencies.

2. **Frontend (React, Next.js)**
   - Multilingual chatbot, symptom form, file upload, emergency button.
   - Sends requests like `POST /api/symptom-entry` or `POST /api/upload/report` → API Gateway.

3. **API Gateway (FastAPI, Python)**
   - Routes requests to AI/ML Core or Databases.
   - Ensures fast and secure communication.

4. **AI/ML Core (PyTorch, Hugging Face, GPT, Whisper)**

    - Symptom triage (Green/Yellow/Red).
    - Lab report analysis & summaries.
    - Specialist AI (Dermatology, Radiology, Cardiology).
    - CT scan reconstruction + voice/text outputs.

5. **Databases & Storage (PostgreSQL, Redis, Vector DB, S3)**
    - Store user data, triage history, reports, scans.
    - Example queries:
        - `INSERT INTO triage_results ...`
        - `SELECT * FROM lab_reports WHERE user_id=?`.

6. **Security**
    - HIPAA/GDPR compliance, JWT auth, SSL encryption.
    - Every API/DB call passes through this layer.

7. **Outputs**
    - Triage result, medical summary, lab report explanation, CT scan reconstruction, govt scheme eligibility, emergency first-aid.
    - Delivered as **Text, Voice, or Infographics**.

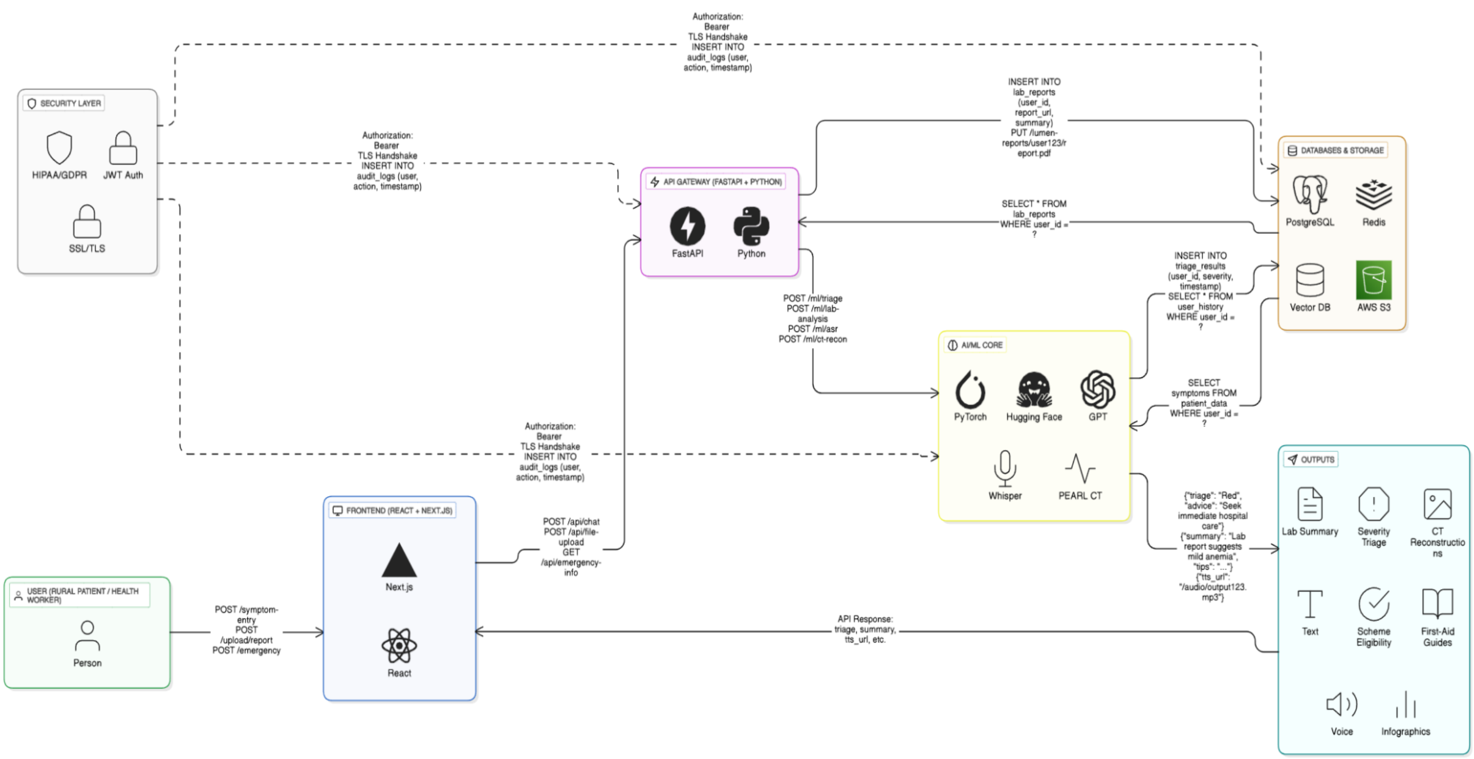

Diagram 2: System architecture diagram

4. Features

Normal Features:

- Symptoms-Based Diagnosis & Guidance

- Accepts multimodal inputs (text / audio / image).
- Analyzes reported symptoms and provides a **probable diagnosis with severity categorization (Green / Yellow / Red)**.
- Offers **clear next-step guidance** (home care, clinic visit, or emergency attention), along with easy-to-follow instructions in **voice, text, and visualformats**.

-AI Specialist Modules

- Dermatology, Radiology, Cardiology, etc., offering diagnostic suggestions from multimodal inputs.
- Provides both **patient-friendly advice** and **clinician-level summaries**.

-Multilingual Voice-First Chatbot

- Supports five Indian languages.
- Uses **Whisper** for ASR and **GPT** for natural, empathetic explanations.

Unique Differentiator features:

-PEARL Integration - Personalized Estimated Anatomic Reconstruction & Lifecare

- A local, hybrid CT reconstruction engine combining **geometry-aware modeling (PerX2CT)**, **diffusion refinement (XctDiff)**, and **NeRF detail polishing (SAX-NeRF)**.
- Generates estimated CT volumes with **voxel-level uncertainty**, enabling safer, lower-dose imaging for follow-up

-Lab Report Analyzer & Follow-Up Generator

- Parses uploaded lab reports (PDF/image), compares values with **age- and sex-specific reference ranges**, flags abnormalities, and generates **simple explanations with diet / lifestyle advice**

  Example:

  Input (Lab Report): 30-year-old female, Hemoglobin **9.8 g/dL** (Normal: 12.0–15.5), RBC **3.6 million/μL** (Normal: 4.2–5.4).

Output (System): *"Your blood count is lower than normal, which may cause tiredness. Eat iron-rich foods such as spinach, dal, jaggery, and vitamin C fruits. Please consult a doctor if you feel weak or if levels fall further."*

Severity tier: Moderate — follow-up within 1–2 weeks

-Government Schemes & Benefits Assistant

- Retrieves up-to-date **national and state health schemes** from a knowledge base and explains eligibility + steps in **local language**
  Example:

  Input (User Query): *"My father in Uttar Pradesh needs dialysis - is there any government help?"*

  Output (System): *"Yes. Under Ayushman Bharat – PMJAY and the UP State Health Scheme, eligible patients get free dialysis at government hospitals. Carry Aadhaar and Ayushman Bharat card to the hospital registration desk. You can also call 14555 for support."*

-Preliminary Triage & Emergency Education

- Offers **step-by-step, life-saving instructions** for common emergencies where delays cost lives - such as **snakebite, drowning, burns, or electric shock**.
- Provides clear **voice, text, and visual guidance** in local language (e.g., CPR steps, immobilization after snakebite, "stop-drop-roll" for fire injuries).

  Example:

  Input: "A child has stopped breathing after drowning."

  Output: *"Call for emergency help immediately. Lay the child flat, check breathing. If not breathing, start CPR: 30 chest compressions, 2 rescue breaths. Continue until medical help arrives."*

-GynaeCare:

A specialized women's health module designed for rural and semi-urban contexts.

- Symptom Screening & Awareness: Private Q&A to flag possible issues like **PCOD/PCOS** (irregular cycles, acne, hormonal imbalance) and Endometriosis (severe pelvic pain, painful periods).
- Guided Next Steps: **Early red-flag alerts** on when to seek medical help, plus low-cost self-care tips (diet, activity, stress management).
- Sanitation & Hygiene Education: Safe use of cloth pads, menstrual cups, biodegradable pads, **with proper disposal methods** (ash pits, eco-friendly options).
- **Govt Schemes & Support**: Links to Jan Aushadhi (affordable medicines), Menstrual Hygiene Scheme (MHS), and connects users with local ASHA/anganwadi workers

5.Role of OpenAI Tools

LUMEN Feature	OpenAI Model	Usecase
Symptom Triage & Guidance	gpt-4o	Provides empathetic triage and severity classification from patient symptoms
AI Specialist Summaries	gpt-4o-mini	Summarizes AI-ML Model outputs into doctor-style report
Lab Report Analyzer	gpt-4o	Interprets OCR lab values and explains results in patient-friendly terms
Govt Schemes Assistant	text-embedding-3-small+gpt-4o-mini	Retrieves and explains govt health scheme eligibility in simple language.
Emergency Protocols	gpt-4o-mini	Gives fast, step-by-step emergency medical instructions.
Voice Input (ASR)	whisper-1	Converts patient speech to text for symptom entry
Voice Output (TTS)	gpt-4o-audio/Azure TTS	Delivers AI responses as a natural voice for accessibility.
Chatbot	gpt-4o	Provides 24/7 conversational support, guiding users across triage, lab results, schemes, and emergencies

6.Tech Stack

LAYER	TECH	USE
Frontend	React (TypeScript), Next.js, Tailwind CSS	Multilingual, responsive web UI for symptom input, lab uploads, CT viewing, chatbot; SSR for speed & SEO
Backend	FastAPI	High-performance backend framework;

		implements REST API endpoints for frontend and AI/ML model communication
Primary DB	PostgreSQL	Stores user profiles, triage history, lab values, CT metadata, government scheme eligibility
Cache	Redis	Session caching, language translation caching
Vector DB	Pinecone	Fully managed vector database for semantic search and embeddings of medical protocols and government schemes
Object Storage	AWS S3	Scalable, secure storage for CT scans, lab reports, medical images; HIPAA/GDPR compliant
AIML core	PyTorch, Hugging Face Transformers, OpenAI APIs (GPT-4o, Whisper, DALL·E)	Hosts AI models (e.g., PEARL CT, dermatology AI) and supports language and vision tasks via OpenAI
Security	JWT, OAuth2	Authentication and authorization mechanisms
Infrastructure & Deploy	Docker, Kubernetes (K8s) on AWS/GCP/Azure, CDN	Containerized deployment, GPU-enabled nodes for AI, CDN for frontend assets delivery

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7.Feasibility

7.1 Technical Feasibility

- **Resources & Technology:**
  - Prototype: Hugging Face free models (Indic-GPT, Donut, Whisper-small).
    Production: OpenAI APIs (GPT-4o, Whisper, DALL·E) + custom PEARL CT pipeline.
- **Infrastructure:**
  - Frontend: React + Tailwind CSS.
  - Backend: FastAPI (Python) with Docker.
  - Deployment: Netlify (frontend), AWS/GCP (production).
- **Assessment:** Existing technologies are sufficient.

Only CT reconstruction pipeline requires GPU resources, which are available on cloud platforms.

7.2 Operational Feasibility

- **Problem Fit:** Addresses rural healthcare gaps (900M+ residents), triage delays, and lab follow-up inefficiencies.
- **Ease of Operation:**
  - Multilingual voice-first chatbot lowers digital literacy barriers.
  - Offline-first design ensures use even in low-connectivity areas.
- **Assessment:** Operationally feasible, since workflows mirror real-world healthcare interactions (symptom → guidance → follow-up).

7.3 Economic Feasibility

- **Prototype Cost:** Minimal (free tiers: Hugging Face, Netlify, Firebase).
- **Production Cost:** API usage (OpenAI GPT, Whisper), GPU compute (CT), and storage (AWS S3).
- **ROI:**
  - Reducing preventable deaths (e.g., 58,000 annual snakebite fatalities).
  - Saving costs from unnecessary clinic visits & repeated CT scans.

- **Assessment:** Strong cost-benefit justification; socially impactful and scalable.

7.4 Legal Feasibility

- **Compliance Requirements:**
  - Data protection → GDPR/HIPAA-like standards.
  - Informed consent → required for data use.
- **Assessment:** Legally feasible with proper compliance in production; no major barriers.

7.5 Market Feasibility

- **Target Users:** 900M+ rural/semi-urban Indians lacking timely healthcare.
- **Market Trend:** Rising smartphone penetration (67%+ rural households with access).
- **Competition:** Existing health apps (Practo, 1mg) focus on urban users; none combine **triage + lab reports + CT + schemes** in one system.
- **Assessment:** High demand, underserved market, unique positioning.

8.Novelty

Feature	Traditional Systems	LUMEN
CT Imaging	Hospital CT scans (₹4,000–₹6,000); no AI low-dose alternatives	PEARL CT: Low-dose AI reconstruction with uncertainty maps → safer & cheaper follow-ups
Lab Report Analysis	1mg, Apollo 24/7 show raw values only	AI Analyzer: Flags abnormalities + gives lifestyle/diet advice in simple local language
Government Schemes Access	info scattered on portals (Ayushman Bharat website, state portals)	Integrated Assistant: Explains eligibility + steps in voice/text for each patient's condition.
Emergency Education	Missing in health apps; patients rely on hearsay or healers	Built-in Protocols: CPR, snakebite, burns → step-by-step local language guidance
Women's Health (LUMEN GynaeCare)	Flo, Clue (cycle tracking); Practo (urban gyne consults); NGOs like Goonj (hygiene awareness). Each addresses only one aspect	integrated GynaeCare: Private symptom screening (PCOD, endometriosis) + hygiene education (safe pad use, disposal) + govt scheme linkage (MHS, Jan Aushadhi) → all in one, voice-first & rural-friendly
Costing	Doctor visit: ₹300–₹500, travel to city hospital: ₹800–₹1,500, CT scan: ₹4,000–₹6,000, follow-ups ~₹1,000	Cuts costs by 50–70% through local AI triage, fewer city visits, and reduced repeat scans

8. Impact & Benefits

Quantitative Benefits

- Reduction in Preventable Morbidity and Mortality: By providing immediate symptom-based triage, emergency education, and guidance, LUMEN aims to significantly reduce the 58,000 annual deaths from snakebites and other emergencies in rural India.
- Cost Savings for Patients: Early and accurate triage can help avoid 20–30% unnecessary hospital visits and repeat CT scans. Since a single CT costs ₹3,000–₹8,000 and a hospital visit costs ₹500–₹2,000, this translates to an average saving of ₹1,200–₹2,800 ($50–$200) per patient.
- Improved Diagnostic Efficiency: Automated analysis of lab reports and specialist modules can reduce diagnostic delays from 2–7 days down to under 1 hour (98% faster turnaround). This efficiency also frees up doctors' time, enabling them to see 2–3 additional patients per hour and reducing complication-related treatment costs by 15–20% in time-sensitive conditions.

Potential Beneficiaries

- Rural and Semi-Urban Populations: Over 900 million residents with limited access to qualified medical professionals.
- Primary Health Centers (PHCs) & Community Health Centers (CHCs): Equipped with decision support for frontline healthcare workers.
- Government Health Schemes Beneficiaries: Increased awareness and access to schemes like Ayushman Bharat, ensuring eligible patients receive entitled benefits.

Awareness & Accessibility Gains

- Multilingual, Voice-First Interface: Supports five Indian languages, enabling accessibility for illiterate or non-English-speaking users.
- Awareness of Government Schemes: Reduces the knowledge gap regarding available health benefits, empowering users to claim entitlements without intermediary assistance.

9. Future Scope

Language Expansion

- Extend support to more Indian regional languages and dialects to further improve inclusivity and reach across diverse linguistic regions of India.

Additional Specialist Modules

- Incorporate more AI-driven modules in fields such as Pediatrics, Obstetrics & Gynecology, Psychiatry, and Neurology for broader diagnostic support and guidance.

NGO & Hospital Integrations

- Collaborate with NGOs and hospitals to integrate LUMEN into field operations, enabling real-time reporting and referrals from remote areas to specialized centers.

Offline-First Mobile App

- Develop a fully-featured Android app with offline-first capabilities, integrating preloaded emergency protocols, government schemes, and first-aid guidance for even deeper rural penetration.

Predictive Healthcare Analytics

- Leverage patient data and interaction history to provide predictive health risk analytics and early warnings for chronic diseases.

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11. References

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